

Operation Covert Access: Weaponized LNK-Based Spear-Phishing Targeting Argentina's Judicial Sector to Deploy a Covert RAT

 seqrite.com/blog/operation-covert-access-weaponized-lnk-based-spear-phishing-targeting-argentinas-judicial-sector-to-deploy-a-covert-rat

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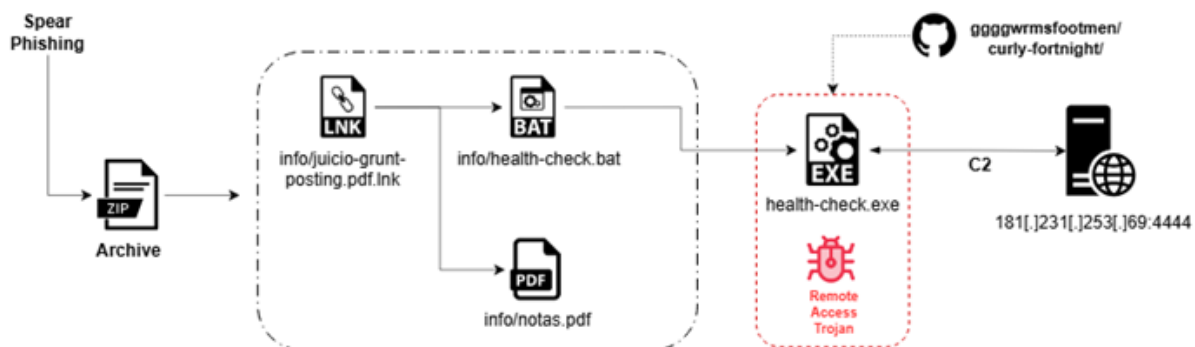
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Introduction:

Seqrite Labs has identified and uncovered a globally active spear-phishing campaign targeting Argentina's judicial sector. The campaign leverages a multi-stage infection chain to deploy a stealthy Remote Access Trojan (RAT), demonstrating a high level of operational sophistication. This spear-phishing campaign abusing legitimate Argentine federal court rulings related to preventive detention reviews to deliver a Rust-based Remote Access Trojan.

Infection Chain:



Targeted Sectors:

The campaign primarily targets Argentina's judicial sector, extending to legal professionals, justice-adjacent government bodies, academic institutions, and legal advocacy organizations. The campaign leverages highly authentic judicial decoy documents to exploit trust in court communications, enabling successful delivery of a

covert Remote Access Trojan and facilitating long-term access to sensitive legal and institutional data.

South America (Primary Focus): Argentina

Judicial institutions, Legal professionals, Justice-adjacent government bodies

Rationale: The decoy documents reference Argentine federal courts and use region-specific legal terminology, indicating focused targeting within Argentina.

Initial Findings about Campaign:

Based on detailed analysis, we assess this activity as a highly targeted, multi-stage intrusion campaign that leverages trusted platforms, advanced anti-analysis techniques, and a covert Rust-based Remote Access Trojan (RAT) to establish persistent access within judicial-sector environments.

The campaign is delivered via targeted spear-phishing emails containing a ZIP archive. The archive includes a weaponized LNK file, a BAT-based loader script, and a legitimate-looking judicial PDF used as a decoy. Upon user interaction with the LNK file, the execution chain is initiated while the decoy document is displayed to the victim, enabling stealthy deployment of the final payload.

Zip: D:\auto_black_abuse\resources\unzipped\20251215_141941_2025-11-28\13adde53bd767d17108786bcc1bc0707c2411a40f11d67dfa9ba1a2c62cc5cf3.zip

- LNK: info/juicio-grunt-posting.pdf.lnk
- BAT: info/health-check.bat
- Decoy/PDF: info/notas.pdf

Analysis of Decoy:

The decoy document used in this campaign is a legitimate-looking Argentine federal court resolution written in formal legal Spanish. It purports to originate from the Poder Judicial de la Nación and references a real judicial body (Tribunal Oral en lo Criminal y Correccional N° 2 de la Capital Federal), complete with case numbering, judicial signatures, and procedural language. The document discusses a judicial review of preventive detention and grants conditional release (excarcelación) to the accused, citing medical evaluations, legal justifications, and specific articles of the Argentine Criminal Procedure Code. Its structure, terminology, and formatting closely mirror authentic court rulings, significantly increasing its credibility among legal and judicial professionals.

Judicial Branch of the Nation
ORAL CRIMINAL AND CORRECTIONAL COURT
NO. 2 OF THE FEDERAL CAPITAL
CCC 45.834/2024/TO1
Buenos Aires, June 19, 2025


Poder Judicial de la Nación
TRIBUNAL ORAL EN LO CRIMINAL Y CORRECCIONAL NRO. 2 DE LA CAPITAL FEDERAL
CCC 45.834/2024/TO1
///nos Aires, 19 de junio de 2025.

AND HAVING SEEN:
The request for review of the detention made by Dr.
Alejandro Argibay Molina, in favor of his client Gaspar
Onesto de Luca -currently detained under house arrest-

Y VISTO:

El pedido de revisión de la detención formulado por el Dr.
Alejandro Argibay Molina, en favor de su asistido Gaspar Onesto de Luca
—actualmente detenido bajo la modalidad de la prisión domiciliaria—.

Y CONSIDERANDO:

I. Que, en la presentación efectuada en el día de la fecha, la
defensa manifestó: "... Si bien no se ha firmado acuerdo alguno con la fiscalía, se le
ha ofrecido a esta parte una posibilidad de firmar un juicio abreviado 3 años de condena
y libertad inmediata con severas reglas de conducta para el imputado.

*Lo cierto es que en este proceso JAMAS, se podrá condenar a GAS-
PAR ONESTO DE LUCA a una pena superior a los 3 años en suspenso, y con
ello bajo alusión a que si bien restan pericias que realizar que darán sustento a este pe-
dido a VVEE, NO MENOS CIERTO ES que no van a existir hasta la fecha de*



Poder Judicial de la Nación

TRIBUNAL ORAL EN LO CRIMINAL Y CORRECCIONAL NRO. 2 DE LA CAPITAL FEDERAL
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The decoy serves as a social-engineering mechanism designed to exploit trust in judicial communications. By embedding the malware delivery chain behind a document that aligns with routine legal workflows—such as detention reviews and court resolutions—the attackers increase the likelihood of user interaction while minimizing suspicion. The document’s relevance to judicial personnel, legal practitioners, and affiliated institutions strongly suggests deliberate, sector-specific targeting rather than opportunistic distribution. The use of an authoritative legal decoy, combined with a covert execution chain, indicates a calculated effort to achieve stealthy initial access and long-term persistence within judicial-sector environments.



Poder Judicial de la Nación

TRIBUNAL ORAL EN LO CRIMINAL Y CORRECCIONAL NRO. 2 DE LA CAPITAL FEDERAL
CCC 45.834/2024/TO1

antecedentes médicos y recomendaciones que se consideren de interés. d) Si se encuentra en condiciones de ser sometido a juicio: Desde el punto de vista pericial psico-psiquiátrico se valora que se encuentra en condiciones biopsíquicas de ser sometido a juicio. e) Si resulta peligroso tanto para sí como para terceros: No se advierten en su examen

Technical Analysis:

We analyzed a downloaded ZIP (D:\auto_black_abuse\resources\unzipped\20251215_141941_2025-11-28\13adde53bd767d17108786bcc1bc0707c2411a40f11d67dfa9ba1a2c62cc5cf3.zip) Archive that contained a malicious LNK (info/juicio-grunt-posting.pdf.lnk) file, a BAT-based loader script (info/health-check.bat), and a judicial-themed PDF decoy document. The archive served as the initial delivery package for the campaign's multi-stage execution chain.

Stage-1: Analysis of Windows Shortcut file (.LNK) – info/juicio-grunt-posting.pdf.lnk:

During our analysis, we extracted the contents of the LNK file with the above-mentioned name, which contained a very simple code as shown below.

```
Windows
System32
WindowsPowerShell
powershell.exe
C:\Windows\System32\WindowsPowerShell\v1.0\powershell.exe
```

grunt-pc
Windows
System32

UWindowsPowerShell

```
powershell.exe *C:\Windows\System32\WindowsPowerShell\v1.0(-ep bypass -w hidden -f health-check.bat%E:\repos\C2R2-v2\dropper\pdf_icon.ico
```

The LNK file is used to stealthily execute a batch script by launching PowerShell from the system directory. It bypasses PowerShell's execution policy and runs in hidden mode to avoid user detection. The inclusion of a PDF icon suggests the shortcut is disguised as a legitimate file, indicating an attempt to deceive users and silently execute the payload.

Stage-2: Analysis of BAT file – info/health-check.bat:

The batch file establishes a connection to a GitHub-hosted URL and retrieves a second-stage payload, which appears to be a Remote Access Trojan (RAT), indicating a multi-stage malware delivery mechanism.

This PowerShell command silently downloads and executes a payload from a GitHub repository by launching PowerShell in hidden mode with execution policy bypassed, thereby avoiding user visibility and security controls. It creates a WebClient object and assigns a legitimate-looking User-Agent string to blend in with normal browser traffic and evade detection mechanisms.

The command retrieves a file named health-check.exe from GitHub and saves it as msedge_proxy.exe within the Microsoft Edge user data directory, using a trusted filename and location to appear legitimate. Finally, it executes the downloaded binary using Start-Process, triggering the payload and indicating a likely second-stage malware execution.

EXE file exhibits environment and virtualization detection behaviour by spawning multiple command-line processes to query specific Windows registry keys. It checks for artifacts related to popular virtualization and sandbox platforms, including VMware, VirtualBox, and Hyper-V, by querying registry paths associated with their tools and services. This activity indicates that the executable is attempting to determine whether it is running inside a virtualized or analysis environment.

Stage-3: Details analysis of RAT – msedge_proxy.exe:

Upon analysis we found that msedge_proxy.exe is performs extensive anti-VM, anti-sandbox, and anti-debug checks, immediately terminating execution if analysis artifacts are detected.

It collects host system information, establishes a resilient C2 connection with IPv4/IPv6 fallback, and advertises a modular command set supporting persistence, file transfer, harvesting, encryption, and privilege escalation. Commands are Base64-encoded, dynamically executed (including DLL-based ransomware and stealer modules), with full lifecycle support for persistence installation and clean removal.

Below is the detail analysis of msedge_proxy.exe.

1. WMIC Manufacturer Check #1

The function begins with:

```
wmic computer system get manufacturer
```

It captures the output → lowercases it → compares against a set of virtualization vendor strings:

- vmware
- virtualbox
- qemu
- microsoft corporation (Hyper-V)
- xen
- parallels

- registry paths belonging to virtual environments

If the output matches → the malware exits immediately.

2. MASSIVE SUBSTRING MATCH: VM / Sandbox Indicators

It loads a very large, concatenated string containing:

Registry Keys:

HKLM\SOFTWARE\VMware, Inc.\VMware Tools
HKLM\SYSTEM\ControlSet001\Services\vmmouse
HKLM\SYSTEM\ControlSet001\Services\vmhgfs
HKLM\SOFTWARE\Oracle\VirtualBox Guest Additions
HKLM\HARDWARE\ACPI\DSDT\VBOX__
HKCU\Software\Wine

VM/Analysis Files:

C:\windows\System32\Drivers\Vmmouse.sys
C:\windows\System32\Drivers\vmhgfs.sys
C:\windows\System32\Drivers\VBoxMouse.sys
C:\windows\System32\Drivers\VBoxGuest.sys
C:\windows\System32\Drivers\VBoxSF.sys
C:\windows\System32\vboxdisp.dll
C:\windows\System32\vboxhook.dll
C:\windows\System32\vboxogl*.dll

Sandbox directories:

C:\analysis
C:\sandbox
C:\sample.exe
C:\malware.exe

VM MAC prefixes:

00:05:69 (VMware)
00:0c:29 (VMware)
00:1c:14 (VMware)
00:50:56 (VMware)
08:00:27 (VirtualBox)
52:54:00 (QEMU/KVM)
00:15:5d (Hyper-V)

Analysis Tools:

procmon.exe
procxp.exe
wireshark.exe
fiddler.exe
ollydbg.exe
ida.exe
ida64.exe
x64dbg.exe
x32dbg.exe
windbg.exe

It iterates through all of these: substring is present → exit program

3. Tasklist Scan: Detect Analysis Processes

It executes:

task list

Then looks for suspicious processes:

- vmsrvc.exe
- vmusrvc.exe
- vboxtray.exe
- vmwaretray.exe
- vmwareuser.exe
- vmacthlp.exe
- sandboxiedcomlaunch.exe
- sandboxierpcss.exe
- procmon.exe
- procexp.exe
- wireshark.exe
- fiddler.exe
- ollydbg.exe
- ida.exe / ida64.exe
- x64dbg.exe / x32dbg.exe
- windbg.exe

If found → exit

4. File Existence Checks

After that it checks around 128 file paths:

```
if Path::exists(path[i]):
    exit()
```

These are the VM driver files, debug tools, sample folder names etc.

This loop performs a simple anti-sandbox / anti-analysis path check: it iterates through a list of hard-coded paths in 16-byte increments and calls `std::path::Path::exists()` on each one. If any of those paths exist—typically directories or files associated with virtual machines, analysis tools, or sandboxes—the condition becomes true, and the malware immediately terminates the process.

Anti-Debugging: PEB Checks:

This code performs multiple anti-debug checks: it first detects debuggers using `IsDebuggerPresent` and the PEB's `BeingDebugged` flag and immediately exits if either is true. It then runs a timing-based anti-analysis test by measuring time before and after a short sleep to spot breakpoint delays or VM slowdown. Finally, it validates the

system's high-resolution performance counter using QueryPerformanceFrequency, and if the timer behaves abnormally—as in emulators or instrumented environments—the malware triggers a failure and terminates.

5. System Information collection:

This function is the malware's system information collector.

It builds the values for:

- hostname
- username
- OS name
- privilege level (Admin/User)
- After determining the field, it executes:
- hostname → PowerShell WMI query
- username → echo %USERNAME% or WMI fallbacks
- OS → WMI ProductName or registry fallback
- privileges → net session technique
(net session >nul 2>&1 && echo Admin || echo User)

6. C2C connection:

After Anti-Vm checks then malware tries to connect network connection routine.

The routine first attempts to parse the C2 address as a standard IPv4 endpoint in the form IP:port, for example 181.231.253.69:4444.

If this IPv4 parsing fails—either due to invalid numeric octets, missing delimiters, or incorrect format—it then falls back to an IPv6 parsing routine that supports bracketed formats such as [xxxx:xxxx:xxxx::1]:port, correctly separating the address block from the port.

When both IPv4 and IPv6 parsing fail, the malware finally defaults to a hardcoded, embedded command-and-control (C2) string to ensure it always has a fallback server to contact.

```
181.231.253.69:4444__PERSIST__:PERSIST_REMOVE____BEACON:DOWNLOAD:UPLOAD|HARVEST"  
"ENCRYPT:DECRYPT:__ELEVATE__\n" "<>\n";
```

This is the default fallback Command-and-Control server.

Whenever parsing fails, the malware reverts to this string.

Perform Initial C2 Handshake

After connecting, the malware must "identify itself" to the server.

This commonly includes:

Sending a BEACON message

Usually contains:

Field	Purpose
Victim ID	Unique host fingerprint
Process Name	Helps operator identify host
Privilege Level	SYSTEM / Administrator / User
OS Version	For compatibility
Architecture	x86 / x64
Current Time	Used for scheduling
Capabilities	What the agent supports

Command Set:

C2 Command	Action
__PERSIST__	Install persistence (task scheduler, startup, etc)
__PERSIST_REMOVE__	Remove persistence
__BEACON__	Re-beacon / heartbeat
__DOWNLOAD__	Download file (C2 → victim)
__UPLOAD__	Upload file (victim → C2)
__HARVEST__	Steal data
__ENCRYPT__	Encrypt file(s)
__DECRYPT__	Decrypt file(s)

__ELEVATE__

Try privilege escalation

These are EXACT command identifiers the malware advertises to C2.

This matches a modular post-exploitation backdoor.

__PERSIST__ command (Persistence & Privilege management)

Registry Run (User):

Adds values under HKCU\Software\Microsoft\Windows\CurrentVersion\Run using reg add arguments constructed via std::process::Command, with randomized “legit-looking” value names such as SecurityHealthSystray, OneDriveSetup, AdobeAAMUpdater, GoogleChromeAutoLaunch, MicrosoftEdgeAutoLaunch, Teams Machine Installer. Error strings include “Error en reg add” (Spanish).

Scheduled Tasks:

Creates tasks via schtasks with /TN, /TR, /DELAY0001:00, optionally /RL HIGHEST. The binary builds argument vectors and issues Command::output, with clean-up messages like “Scheduled Tasks limpiadas”.

PERSIST_REMOVE (Remove persistence):

After implementing multiple persistence mechanisms (Registry Run, Scheduled Tasks, and WMI Subscriptions), this Rust-written malware also includes a complete cleanup capability.

This is executed when the C2 sends the command:

PERSIST_REMOVE__BEACON:DOWNLOAD:UPLOAD|HARVEST__ENCRYPT:DECRYPT:ELEVATE <>

Registry cleanup: Deletes all Run key entries it previously created.

Scheduled Task removal: Eliminates all tasks associated with its persistence.

BEACON (Re-beacon / heartbeat):

The BEACON command is central to maintaining communication between an infected host and the attacker's command-and-control (C2).

The code snippet below represents the malware's logic for handling the command:

```
__BEACON__:__DOWNLOAD__:__UPLOAD__|__HARVEST____ENCRYPT__:__DECRYPT__:__ELEVATE__  
<<END>>
```

This command instructs the implant to:

Send a heartbeat / beacon back to the server

Include device info

Optionally prepare for further commands

Stay alive and quietly maintain persistence

__DOWNLOAD__:

__DOWNLOAD__ command is not about fetching files from the C2 server. Instead, it's a file exfiltration mechanism—stealing files from the victim and sending them to the attacker. This naming trick is common in RATs and stealers.

This is the most important stage.

The malware builds a formatted message containing:

File name

File size

Base64 data

The formatting structure: [filename | size | base64-data]

UPLOAD:

This malware also provides a full-featured file upload module, triggered by the command:

```
__UPLOAD__|__HARVEST____ENCRYPT__:__DECRYPT__:__ELEVATE__  
<<END>>
```

This allows the attacker to send arbitrary files to the victim machine.

This capability is required for:

Dropping additional payloads

Updating implants

Deploying ransomware modules

Side-loading DLLs

Executing follow-up stages

HARVEST:

The code block triggers when the server sends:

```
__HARVEST__ __ENCRYPT__:__DECRYPT__:__ELEVATE__  
<<END>>
```

This command activates a dormant credential-harvesting module stored on disk as encrypted files.

The malware expects two files to already exist on the victim:

stealer.enc → encrypted DLL

stealer.key → XOR key

If either is missing, the malware replies:

__ERROR__: stealer.enc no encontrado. El servidor debe subirlo primero.

__ERROR__: stealer.key no encontrado. El servidor debe subirlo primero.

ENCRYPT + DECRYPT commands:

The malware receives C2 commands like __ENCRYPT__:__DECRYPT__:__ELEVATE__ or __DECRYPT__:__ELEVATE__ and parses the target folder + mode.

It checks for ransomware.enc (encrypted DLL) and ransomware. Key, then decrypts the DLL using the provided key.

It loads the DLL dynamically and calls either encrypt_directory or decrypt_directory based on the command.

The ransomware engine processes all files in the specified path and returns a text result (OK, error, or encoded data).

__ELEVATE__ command:

The complete implementation of the malware's __ELEVATE__ command—the privilege-escalation module. It generates a PowerShell Elevation Script, writes it to disk, runs it with RunAs, waits for UAC elevation, deletes the script, and returns a status message to the C2.

7 .COMMAND RECEIVING & DECODING:

Commands arrive encoded in BASE64, processed by:

```
agent::base64_decode::_$u7b$$u7b$closure$u7d$$u7d$:h968157c94086d467
```

The decoding procedure decodes 4-byte sequences at a time, reconstructs original message:

3 bytes output per 4 bytes input

Handles padding '=' cases

Pushes decoded byte into vector structures

This proves C2 commands are always base64-encoded.

Conclusion:

We named this campaign Operation Covert Access to reflect its emphasis on stealthy initial access and persistent remote control within judicial-sector environments, rather than reliance on any single malware family or tooling.

Operation Covert Access demonstrates how judicial-themed spear-phishing, combined with weaponized LNK files and a stealthy Remote Access Trojan (RAT), can be leveraged to establish long-term access within high-trust institutional environments. The campaign highlights the continued effectiveness of shortcut-based execution and underscores the need for enhanced visibility and defensive controls against socially engineered intrusion chains.

Seqrite Coverage:

- Trojan.50322.SL
- Trojan.50321.SL
- *CovertRATCiR*

IOCs:

Hash	Name
dc802b8c117a48520a01c98c6c9587b5	info/juicio-grunt-posting.pdf.lnk
45f2a677b3bf994a8f771e611bb29f4f	D:\auto_black_abuse\resources\unzipped\20251215_140518_2025-11-28\13adde53bd767d17108786bcc1bc0707c2411a40f11d67dfa9ba1a2c62cc5cf3.zip
02f85c386f67fac09629ebe5684f7fa0	info/health-check.bat
976b6fce10456f0be6409ff724d7933b	\msedge_proxy.exe
233a9dbcfe4ae348c0c7f4c2defd1ea5	info/notas.pdf

C2

181.231.253.69:4444

MITRE ATT&CK:

Stage	Technique Name	Technique ID
Initial Access	Spear phishing Attachment	T1566.001
	User Execution: Malicious File	T1204.002
	Archive via Email	T1566.001

Execution	Windows Command Shell	T1059.003
	PowerShell	T1059.001
	Command-Line Interface	T1059
	Native API	T1106
	Scheduled Task Execution	T1053.005
Defense Evasion	Masquerading (LNK as PDF)	T1036.004
	Masquerading (Legitimate Name or Location)	T1036.005
	Hidden Window	T1564.003
	Obfuscated / Encoded Commands (Base64)	T1027
	Deobfuscate / Decode Files or Information	T1140
	Execution Policy Bypass	T1562.001
	Modify Registry	T1112
	Virtualization / Sandbox Evasion	T1497
	Virtualization / Sandbox Evasion: System Checks	T1497.001
	Virtualization / Sandbox Evasion: User Activity Checks	T1497.002
	Debugger Evasion	T1622
	Indicator Removal on Host	T1070
Discovery	System Information Discovery	T1082
	Account Discovery	T1087
	Process Discovery	T1057
	Registry Discovery	T1012
	File and Directory Discovery	T1083
	Software Discovery	T1518
	Permission Groups Discovery	T1069
Credential Access	Credentials from Password Stores	T1555
	OS Credential Dumping	T1003
Persistence	Registry Run Keys / Startup Folder	T1547.001
	Scheduled Task / Job	T1053.005
	Boot or Logon Autostart Execution	T1547
	WMI Event Subscription	T1546.003

Command and Control	Application Layer Protocol	T1071
	Application Layer Protocol: Web Protocols	T1071.001
	Encrypted Channel	T1573
	Encrypted Channel: Symmetric Cryptography	T1573.001
	Obfuscated / Encoded Channel	T1132
	Fallback Channels	T1008
	Non-Standard Port	T1571
	Dynamic Resolution (IPv4/IPv6 Parsing)	T1568
	Ingress Tool Transfer	T1105
Collection	Data from Local System	T1005
	Archive Collected Data	T1560
	Input Capture	T1056
	Screen Capture	T1113
Exfiltration	Exfiltration Over C2 Channel	T1041
	Exfiltration Over Encrypted Channel	T1041
Impact	Data Encryption for Impact	T1486
	Inhibit System Recovery	T1490
Lateral Movement	Remote Services	T1021

Authors:

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He is a Security Researcher at Quick Heal Technologies Ltd, working with the Seqrite Lab team. His areas of expertise include threat hunting, malware research and...

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